

## Original articles

# Three-dimensional D3Q27 multiple-relaxation-time lattice Boltzmann simulation of Herschel–Bulkley viscoelastic fluids in a cubic cavity with top lid driven diagonally

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## ABSTRACT

Graphics Processing Unit (GPU) accelerated multiple-relaxation-time (MRT) lattice Boltzmann method (LBM) is used for the simulation of Herschel–Bulkley non-Newtonian fluids in a three-dimensional (3D) cubic cavity with the top lid-driven diagonally. For the 3D simulation, a D3Q27 lattices model, which is more stable and well-accepted in the LBM community, is used in the present MRT-LBM. Simulations using numerical models are run for a variety of dimensionless variables, including the Reynolds numbers ( $Re = 300, 600, 1000, 1200$ ), Bingham number ( $Bn = 0.0, 0.5, 1.0, 2.0$ ), Power-law index, ( $n = 0.8$ ). In the present numerical simulation, the GPU has used a parallel computing technique based on the Compute Unified Device Architecture (CUDA) C++ programming. MRT-LBM code is validated for the Newtonian and non-Newtonian power law fluid with a lid-driven cubic cavity. The numerical results obtained regarding the streamlines, velocity, viscosity distributions, and the iso-surfaces of the non-Newtonian viscosity are presented. The current numerical findings could potentially function as benchmark results for validating 3D codes validation for the non-Newtonian fluids.

## 1. Introduction

The lid-driven cavity flow serves as a fundamental computational benchmark extensively explored by researchers due to its diverse range of flow phenomena across varying Reynolds numbers and flow characteristics. Numerous investigations have employed the lattice Boltzmann method (LBM) to conduct in-depth analyses of three-dimensional (3D) lid-driven cavity flows. Gonzalez et al. [1] delved into the instability of 3D laminar two-dimensional states within a lid-driven isosceles triangular cavity, revealing consistent flow patterns and wave numbers of steady and critical modes. Albensoeder et al. [2] examined centrifugal-flow instabilities in three-dimensional lid-driven cavities, noting the formation of strong eddies downstream in shallow cavities as Reynolds numbers increase. Iwatsu et al. [3] investigated the numerical simulation of three-dimensional flow structures in driven cavities, concluding that low Reynolds numbers primarily affect the fluid near the bottom cavity. In contrast, higher Reynolds numbers yield global structures resembling classical boundary layer flow. Zhong et al. [4] presented a unified incompressible lattice BGK model applied to three-dimensional lid-driven cavity flow, demonstrating stability and accuracy even at high  $\omega$  values, which typically cause instability in other boundary condition schemes. Bouffanaïs et al. [5] analyzed large-eddy flow simulations in a lid-driven cubical cavity, identifying the downstream-corner-eddy region above the bottom wall as the primary site of turbulence production. Safdari and Kim [6] investigated the three-dimensional behavior of particles with varying density ratios in lid-driven cavity flow using LBM.

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Their findings revealed that lighter particles tended to split symmetrically as they approached boundaries, while heavier particles exhibited maximum sedimentation velocity within seconds.

Batol and colleagues [7] utilized the lattice Boltzmann Two-Relaxation-Time scheme to analyze and numerically explore advective and dispersive transport in Herschel-Bulkley fluids. Their findings revealed that while the effective dispersion coefficient relies on rheological parameters, the time required to enter the dispersive regime remains largely independent of fluid rheology. Wu et al. [8] investigated an enhanced MRT-LBM for Herschel-Bulkley fluids at high Reynolds numbers, observing the effectiveness of their proposed method in achieving stable streamlines with increasing  $Re$ . The convection heat transfer within a lid-driven cavity containing a shear-thinning Herschel-Bulkley fluid was examined by Labsi et al. [9]. They concluded that the number of unyielding areas within the enclosure increases with the rise of the generalized Bingham number. Sahu and Matar [10] analyzed three-dimensional linear instability in pressure-driven two-layer channel flow of both Newtonian and Herschel-Bulkley fluids. Their study revealed the presence of three-dimensional instabilities when the square root of the viscosity ratio exceeds the thickness ratio of the two layers. In the investigation by Samantaray et al. [11], turbulence characteristics of high Reynolds number flow within a three-dimensional cubic lid-driven cavity were explored. They noted that the integrated value of helicity density approaches zero, suggesting a predominant orthogonality between velocity and vorticity vectors, with vortex filaments appearing unknotted or unlinked. The regularized lattice Bhatnagar-Gross-Krook model for two- and three-dimensional cavity flow simulations were examined by Montessori et al. [12]. They discovered that the regularized BGK scheme significantly improves the convergence and stability of the standard LBGK model for the case of two- and three-dimensional lid-driven cavity flows. Montessori et al. [13] developed a thread-safe lattice Boltzmann model for high-performance turbulent flow simulations. They found that their results show excellent performance with the coding simplicity of standard lattice Boltzmann implementations, a significantly lower memory footprint than fused streaming-collision strategies, and no need for more complex and exotic streaming algorithms. The use of thread-safe lattice Boltzmann for high-performance computing on GPUs was investigated by Montessori et al. [14]. Thread-safe lattice Boltzmann decreases memory footprint while keeping traditional LB computation's algorithmic simplicity and reaching peak performance in two and three dimensions.

An MRT-LBM model has been presented by Chai et al. [15] for generalized Newtonian fluid flows. The numerical analysis showed that the proposed MRT model was a second-order scheme with better stability than the BGK model. Perumal [16] used the lattice Boltzmann method to simulate the laminar flow in a three-dimensional two-sided lid-driven cubical cavity. It was found that, for 3D parallel wall motion, a pair of counter-rotating secondary vortices is formed along with two primary vortices. On the contrary, for anti-parallel motion, only a single primary vortex appeared at the center of the cavity. Tang et al. [17] developed a Lattice Boltzmann Model based on D3Q19 on non-uniform meshes combined with curvilinear coordinate to investigate three-dimensional vortex flows in a lid-driven cavity numerically. The simulations successfully predicted the flow fields and the results agreed with the existing literature. Chang et al. [18] simulated three-dimensional deep cavities with various depth-width aspect ratios using D3Q19 multi relaxation time lattice Boltzmann method on graphic processing units. Results showed that the speed of the performance was significantly dependent on the size of the problem and adopted precision. Using MRT-LBM, Li et al. [19] examined such fluid in a square cavity. The investigation outcomes showed complex phenomena of non-Newtonian fluid flow and its effects on the lid-driven cavity flows along with comparative analysis of Newtonian and non-Newtonian fluids. Dong et al. [20] simulated the viscous flow in a three-dimensional lid-driven cavity under different inflow angles using D3Q19 LBM with on-site boundary conditions. The study concluded that both the Reynolds number and the inflow angle affect the flow pattern of a lid-driven cavity. Three-dimensional central moment Lattice Boltzmann Method in the presence of forcing terms has been analyzed by Premnath and Banerjee [21]. It was established that the developed method maintained second-order grid convergence under diffusive scaling. The flow pattern in a 3D lid-driven cavity depends on both the inflow angle and the Reynolds number, according to research done by Dong et al. [20], who used the LBM to simulate viscous flow in the cavity. Using MRT-LBM, Samanta et al. [22] analyzed the transport phenomena in a differentially heated lid-driven cavity. They discovered that at a given value of  $Ri$ , the average value of  $Nu$  increases with  $Pr$  and  $Re$ . Numerical investigation of four-lid-driven cavity flow bifurcation using the MRT-LB method studied by Zhuo et al. [23]. The solution, which appears stable due to the rapid drop of  $\epsilon$  to below  $10^{-14}$  and slight fluctuations around this value, is considered stable. Bhopalam et al. [24] investigated three-dimensional simulations of fluid flows in oscillating lid-driven cavities using the lattice Boltzmann method. The simulations showed that the fluid penetration depth inside the cavity decreased as the oscillating frequency increased.

Devakar et al. [25] examined the pressure-driven fully developed steady flow of non-Newtonian fluids through a porous medium in a straight uniform square duct. It was observed that the velocity and volume flow rate decreased for a couple of stress and porosity parameters. At the same time, it increased in the case of the Jeffrey parameter and pressure gradient. Perumal and Dass [26] examined lattice Boltzmann simulation of two- and three-dimensional incompressible thermal flows. For a range of Rayleigh numbers, the current square cavity LBM results closely matched those in the literature and the IEDDF model, suggesting that LBM could compete with finite difference and finite volume methods even for free convection flows. Adam et al. [27] developed a new three-dimensional cascaded LBM on the D3Q19 lattice to simulate non-Newtonian power-law fluids. The model's performance showed superior numerical stability when compared to the SRT-LBM. Thus, it was concluded that this model could affect complex 3D non-Newtonian fluid flows and that promising outcomes could be achieved. Bisht and Patil [28] examined two-dimensional power-law modeled non-Newtonian fluid flow in driven cavities with undulated bottom walls using MRT-LBM. It was observed that, for shear-thinning fluids, the flow structure at steady-state was complex with the augmentation of the Reynolds number. Using the Lattice Boltzmann Method, Safdari and Kim [29] examined how solid particle behavior could be simulated in a lid-driven cavity flow. They concluded that the cavity's diameter, density, and vortex behavior significantly impact the particle trajectories. Convective heat transfer of power-law non-Newtonian fluids had been studied by Chen et al. [30] from an oscillating hot impeller to cold tank walls

using LBM. The study found that increasing the impeller's aspect ratio and oscillation amplitude increased the heat transfer rate. At the same time, the fluid's power-law index showed an inverse relationship with heat transfer effects on tank walls. Non-Newtonian laminar Fluid flow has been simulated by Thohura et al. [31] in a Lid-Driven Skewed Cavity using power-law viscosity model. The finite volume method discretized the transformed equations and the collocated grid arrangement. The investigation outcomes provided benchmark data for skewed lid-driven cavity flow in case of future references. The mixed convection and sensitivity analysis of engine oil impinging jet flow on a revolving heated cylinder with high Prandtl numbers were investigated by Islam and Molla [32]. They discovered that an increase in the local Nusselt number is correlated with increased Reynolds and Prandtl numbers. Using the log-conformation reformulation in OpenFOAM, Habla et al. [33] investigated the viscoelastic flow numerically in a three-dimensional lid-driven cavity. They concluded that the influence of elasticity on the flow rheology becomes more pronounced as the Weissenberg number increases. The flow of power-law fluids in a cavity driven by the motion of two facing lids was studied by Mendu et al. [34] using the LBM simulation. They concluded that at lower values of  $n$ , the center of primary vortices moves to the upper and lower corners of the cavity in the direction of lid motion due to the fluid's decreasing viscosity.

The novelty of this study lies in its utilization of GPU acceleration to implement a D3Q27 multiple-relaxation-time lattice Boltzmann simulation, specifically tailored for investigating Herschel-Bulkley non-Newtonian fluids within a cubic cavity with a diagonally driven top lid. This approach enables unprecedented computational efficiency and accuracy in capturing complex non-Newtonian fluids' intricate flow behaviors and rheological characteristics under such dynamic conditions, thereby opening new avenues for in-depth exploration and understanding of these systems.

## 2. Herschel-Bulkley non-Newtonian fluids

In the present numerical simulation, a three-dimensional (3D) incompressible Herschel-Bulkley non-Newtonian fluids have been simulated using the D3Q27 MRT-lattice Boltzmann method. The Herschel-Bulkley fluids are a non-Newtonian fluid exhibiting a rheological behavior characterized by a yield stress, which means that they do not flow until a certain stress threshold, known as the yield stress, is exceeded. Once this threshold is surpassed, they flow like a viscous fluid. This behavior is described by the Herschel-Bulkley rheological model, which extends the concept of Newtonian fluids to include materials that exhibit both viscous and elastic properties. The following equation expresses the Herschel-Bulkley model with the Papanastasiou [35] regularization technique:

$$\bar{\tau}(|\dot{\gamma}|) = \left[ K \cdot |\dot{\gamma}|^{n-1} + \frac{\tau_y}{|\dot{\gamma}|} (1 - \exp(-m|\dot{\gamma}|)) \right] \dot{\gamma} \quad (1)$$

where  $\tau$  is the shear stress,  $\gamma$  is the shear-rate,  $K$  is the consistency index,  $n$  is the flow behavior index,  $\tau_y$  is the yield shear stress,  $m$  is the growth factor and  $|\dot{\gamma}|$  denotes the magnitude of the shear-rate as follows:

$$|\dot{\gamma}| = \sqrt{2\gamma_{ij}\gamma_{ij}} \quad (2)$$

where  $\gamma_{ij} \stackrel{\text{def}}{=} \frac{1}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$  is the shear-rate or strain rate tensor.

The non-Newtonian viscosity of the Herschel-Bulkley fluid represents by

$$\mu(|\dot{\gamma}|) = K \cdot |\dot{\gamma}|^{n-1} + \frac{\tau_y}{|\dot{\gamma}|} (1 - \exp(-m|\dot{\gamma}|)) \quad (3)$$

From the viscosity relation (3), it is seen that the combination of the power-law fluid and Bingham fluid models constructs the Herschel-Bulkley model. For  $n = 1$ , the viscosity model (3) represents only for the Bingham fluid and if  $\tau_y = 0$  the model depicts only for the power-law fluids.

This Herschel-Bulkley model is commonly used to describe the flow behavior of various non-Newtonian fluids such as pastes, gels, slurries, and certain types of biological fluids. The behavior of Herschel-Bulkley fluids is crucial in various industrial processes, including food processing, cosmetics manufacturing, oil drilling, and pharmaceutical production, among others. Understanding their rheological properties is essential for optimizing processing conditions and ensuring product quality.

## 3. Computational model and the governing Navier–Stokes equation

In the context of a three-dimensional laminar and incompressible flow involving a non-Newtonian fluid confined within a cubic cavity of height  $H$ , the physical representation of the geometry is depicted in Fig. 1. The flow is initiated by the sliding movement of the top wall along the diagonal of the  $xz$ -plane, and the conservation equations for mass and momentum govern it. In the case of non-Newtonian fluid flow, these equations can be expressed dimensionless.

$$\nabla \cdot \mathbf{u} = 0 \quad (4)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \nabla \cdot \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla \cdot \boldsymbol{\tau} \quad (5)$$

where  $\mathbf{u} = (u, v, w)$  and  $Re \stackrel{\text{def}}{=} \frac{\rho U_0^{2-n} H^n}{K}$  depends on power-law index  $n$ , here  $U_0 = 0.1$  is the velocity magnitude of top-lid sliding velocity and the sliding velocity components are  $(u_l, v_l, w_l) = (\sqrt{2}, 0, \sqrt{2})/20$ .

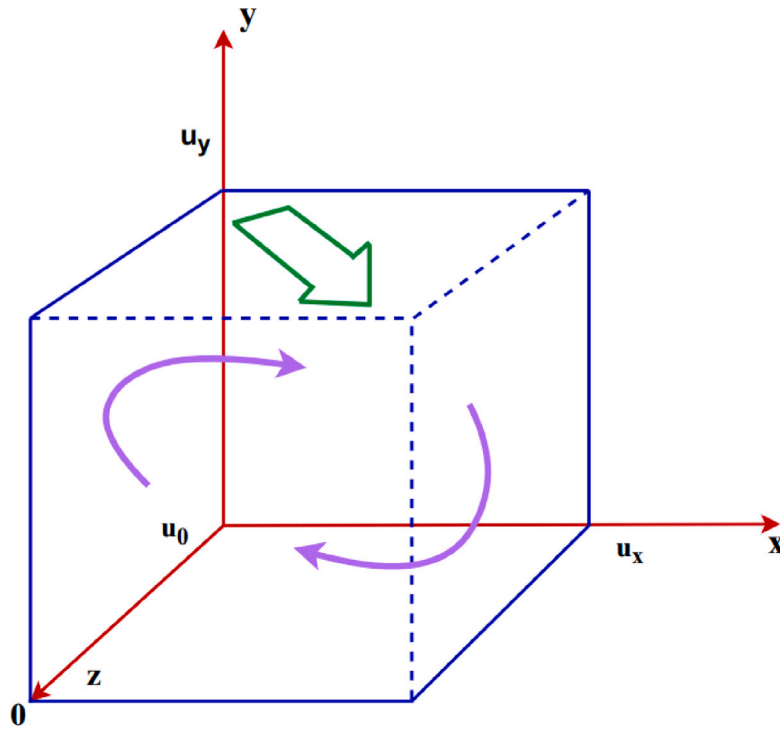


Fig. 1. Schematic model and coordinate systems.

The non-dimensional form of the Herschel-Bulkley model is written as:

$$\tau(|\dot{\gamma}|) = \left[ |\dot{\gamma}|^{n-1} + \frac{Bn}{|\dot{\gamma}|} (1 - \exp(-M|\dot{\gamma}|)) \right] \dot{\gamma} \quad (6)$$

where  $Bn = \frac{\tau_y H}{\frac{\kappa}{\rho} U_0}$  is the Bingham number and  $M = \frac{m U_0}{H}$  is the growth factor which is taken 1000 for the whole simulation.

#### 4. Formulation of the problem in D3Q27 MRT-LBM

In this current numerical simulation, we employ a D3Q27 MRT lattice Boltzmann method, chosen for its numerical stability. The employment of Multiple Relaxation Time (MRT) helps mitigate inconsistencies in numerical solutions arising from the twenty-seven relaxation parameters, as detailed by Suga et al. [36]. The MRT lattice Boltzmann expression for fluid flow operates within the moment space rather than velocity, as presented below:

$$\mathcal{F}(\mathbf{r} + \mathbf{q}_i \Delta t, t + \Delta t) - \mathcal{F}(\mathbf{r}, t) = -\mathbf{M}^{-1} \mathbf{S}_t [\mathbf{m}(\mathbf{r}, t) - \mathbf{m}^{eq}(\mathbf{r}, t)] \quad (7)$$

where  $\mathbf{r} = (\xi, \eta, \zeta)$ . The standard settings for the lattice spacing  $\Delta\xi = \Delta\eta = \Delta\zeta$ , time step  $\Delta t$ , and lattice speed  $c = \Delta\xi/\Delta t = 1$ . The discrete velocities are  $\mathbf{q}_i = [q_{i\xi}, q_{i\eta}, q_{i\zeta}]$  are shown in Fig. 2 and defined as follows:

$$|q_{i\xi}\rangle = \begin{bmatrix} 0 & 1 & 0 & -1 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \end{bmatrix}^T \quad (8)$$

$$|q_{i\eta}\rangle = \begin{bmatrix} 0 & 0 & 1 & 0 & -1 & 0 & 0 & 1 & 1 & -1 & -1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 0 & 1 \end{bmatrix}^T \quad (9)$$

$$|q_{i\zeta}\rangle = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & 1 & 1 \end{bmatrix}^T, \quad (10)$$

Here,  $\mathcal{F} = (f_0, f_1, f_2, \dots, f_{26})$  represents probability distribution functions, while  $\mathbf{m} = (m_0, m_1, m_2, \dots, m_{26})$  denotes velocity moments. Similarly,  $\mathbf{m}^{eq} = (m_0^{eq}, m_1^{eq}, m_2^{eq}, \dots, m_{26}^{eq})$  stands for equilibrium moments pertaining to velocity. Through linear transformation, the mapping between velocity and moment spaces can be realized as  $\mathbf{m} = \mathbf{M}\mathcal{F}$ ,  $\mathcal{F} = \mathbf{M}^{-1}\mathbf{m}$ , and  $\mathbf{m}^{eq} = \mathbf{M}\mathcal{F}^{eq}$ . Here,  $\mathcal{F}_i^{(eq)}$  represents the equilibrium distribution function as described in [37–39].

$$\mathcal{F}_i^{eq} = w_i \rho \left[ 1 + \frac{3(\mathbf{q}_i \cdot \mathbf{u})}{c^2} + \frac{9(\mathbf{q}_i \cdot \mathbf{u})^2}{2c^4} - \frac{3(\mathbf{u} \cdot \mathbf{u})}{2c^2} \right] \quad (11)$$

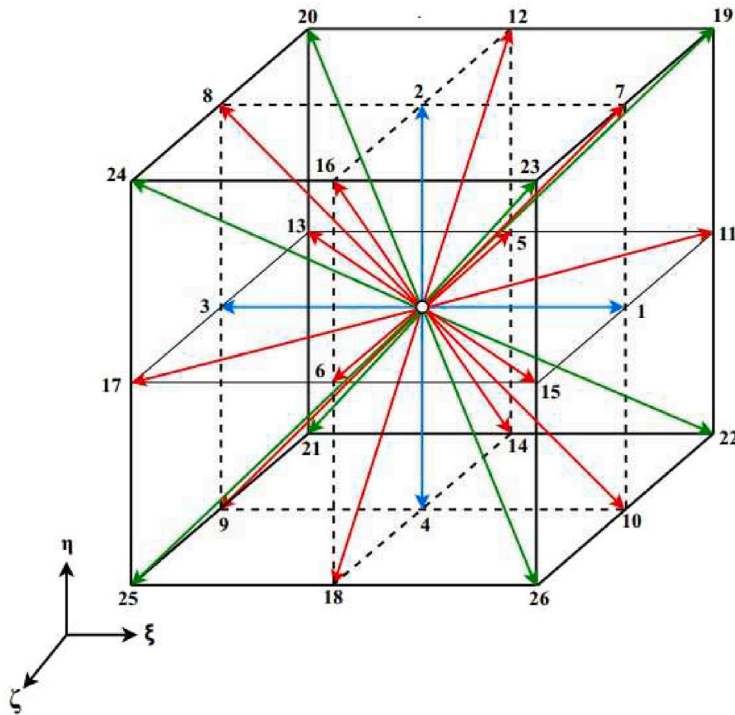


Fig. 2. D3Q27 lattice model.

In Eq. (11),  $w_i$  are  $w_0 = \frac{8}{27}$ ,  $w_{1-6} = \frac{2}{27}$ ,  $w_{7-18} = \frac{1}{54}$  and  $w_{19-26} = \frac{1}{216}$  for the D3Q27 lattice model.

For the D3Q27 lattice model  $\mathbf{M}$  is the  $27 \times 27$  orthogonal transformation matrix. Initially, the non-orthogonal raw-moments are calculated based on the procedure of Dubois and Lallemand [40] as follows:

(i) the density

$$\hat{m}_0 = \rho = \sum_{i=0}^{26} f_i \quad (12)$$

(ii) the momentum

$$\hat{m}_1 = j_x = \rho u = \sum_{i=0}^{26} f_i q_{i\xi}, \quad \hat{m}_2 = j_y = \rho v = \sum_{i=0}^{26} f_i q_{i\eta}, \quad \hat{m}_3 = j_z = \rho w = \sum_{i=0}^{26} f_i q_{i\zeta} \quad (13)$$

Second and higher orders moments are described in Suga et al. [36].

By using the non-orthogonal moments, one can construct the non-orthogonal  $27 \times 27$  matrix  $\hat{\mathbf{M}}$ , that can be normalized by the Gram–Schmidt procedure as follows:

$$\mathbf{M}_j = \hat{\mathbf{M}}_j - \sum_{i=0}^{j-1} \frac{\hat{\mathbf{M}}_j \cdot \mathbf{M}_i}{|\mathbf{M}_i|^2} \mathbf{M}_i \quad (j \geq 4) \quad (14)$$

where  $\hat{\mathbf{M}}_i$  and  $\mathbf{M}_i$  are the  $i$ th rows of  $\hat{\mathbf{M}}$  and  $\mathbf{M}$  matrices, respectively.

In Eq. (7), the collision matrix  $\mathbf{S}_f$  is a  $27 \times 27$  diagonal matrix whose elements are the relaxation parameters as in Suga et al. [36]. The collision matrix  $\mathbf{S}_f$  is written as

$$\mathbf{S}_f = \text{diag}(0, 0, 0, 0, s_4, s_5, s_6, s_7, s_8, s_9, s_{10}, s_{11}, s_{12}, s_{13}, s_{14}, s_{15}, s_{16}, s_{17}, s_{18}, s_{19}, s_{20}, s_{21}, s_{22}, s_{23}, s_{24}, s_{25}, s_{26}) \quad (15)$$

where  $s_4 = 1.54$ ,  $s_5 = s_6 = s_7 = s_8 = s_9 = \frac{1}{(3\nu(\mathbf{r}, t) + 0.5)}$ ,  $s_{10} = s_{11} = s_{12} = 1.5$ ,  $s_{13} = s_{14} = s_{15} = 1.83$ ,  $s_{16} = 1.4$ ,  $s_{17} = 1.61$ ,  $s_{18} = s_{19} = s_{20} = s_{21} = s_{22} = 1.4$  and  $s_{23} = s_{24} = s_{25} = s_{26} = 1.74$ . Here  $\nu(\mathbf{r}, t)$  is the kinematic viscosity of the power-law non-Newtonian fluid which will be calculated locally later on.

The local calculation of non-Newtonian fluid viscosity  $\nu(\mathbf{r}, t)$ , determined by the shear rate, is performed at each node as depicted in Eq. (16), following the methodology outlined in [34].

$$\gamma_{\alpha\beta} = -\frac{3}{2\rho c^2 \tau(\mathbf{r}, t)} \sum_{i=0}^{26} q_{i\xi} q_{i\eta} q_{i\zeta} (f_i - \mathcal{F}_i^{eq}) \quad (16)$$

Chai et al. [15,19] used another relation for the MRT-LBM to calculate the shear-rate for the non-Newtonian fluids as follows:

$$\gamma_{\alpha\beta} = -\frac{3}{2\rho c^2} \sum_{i=0}^8 e_{i\alpha} e_{i\beta} \sum_{j=0}^8 (\mathbf{M}^{-1} \mathbf{S} \mathbf{M})_{ij} (f_j - \mathcal{F}_j^{(eq)}) \quad (17)$$

Here  $f_{\alpha}^{(eq)}$  is calculated from Eq. (11). The above two relations, have been used in the previous studies [41,42] for the D2Q9 and D3Q19 LBM models. For the D3Q27 MRT-LBM, Zhang et al. [43] proposed new relations for the shear-rate calculations as follows:

$$\gamma_{\xi\xi} = -\frac{1}{2\rho} [s_4 m_4^{(1)} + s_5 m_5^{(1)}] \quad (18)$$

$$\gamma_{\eta\eta} = -\frac{1}{4\rho} [2s_4 m_4^{(1)} - s_5 m_5^{(1)} - 3m_6^{(1)}] \quad (19)$$

$$\gamma_{\zeta\zeta} = -\frac{1}{4\rho} [2s_4 m_4^{(1)} - s_5 m_5^{(1)} + 3m_6^{(1)}] \quad (20)$$

$$\gamma_{\xi\eta, \eta\xi, \zeta\xi} = -\frac{3s_7}{2\rho} m^{(1)}_{7,8,9} \quad (21)$$

In the present D3Q27 MRT-LBM, following Zhang et al. [43], the relations (18)–(21) are used in the formula (3) for calculating the non-Newtonian viscosity.

#### 4.1. Mesoscopic boundary conditions

Boundary conditions (BCs) are pivotal for ensuring the stability and accuracy of numerical solutions. Within the framework of Lattice Boltzmann Method (LBM) modeling, careful consideration is necessary to manage discrete distribution functions at boundaries to faithfully represent the macroscopic boundary conditions of the fluid. In this investigation, the bounce-back condition is implemented at the wall. While in macroscopic computational fluid dynamics, the no-slip boundary condition is utilized at walls, in LBM, a mesoscopic bounce-back boundary condition is adopted at the walls. The bounce-back condition stipulates that when a particle from a specific fluid's discrete distribution function reaches a boundary node, it rebounds into the fluid along its incoming direction. In a 3D domain, bounce-back conditions are applied at six surfaces: east, west, north, south, top, and bottom.

Top lid moving diagonally with the velocity  $(u_l, v_l, w_l) = (\sqrt{2}, 0, \sqrt{2})/20$ :

$$\begin{aligned} f_4 &= f_2 \\ f_9 &= f_7 - \frac{u_l}{9} \\ f_{10} &= f_8 + \frac{u_l}{9} \\ f_{18} &= f_{12} - \frac{w_l}{9} \\ f_{14} &= f_{16} + \frac{w_l}{9} \\ f_{21} &= f_{23} - 6 \frac{u_l - w_l}{216} \\ f_{19} &= f_{25} - 6 \frac{u_l + w_l}{216} \\ f_{23} &= f_{21} - 6 \frac{-u_l + w_l}{216} \\ f_{20} &= f_{26} - 6 \frac{-u_l - w_l}{216} \end{aligned} \quad (22)$$

#### 4.2. CUDA C++ programming and performance

Our investigation revealed that the performance of GPU-based lattice Boltzmann method (LBM) simulations surpasses that of CPU-based approaches [44]. Molla et al. [44] extensively elucidated the Compute Unified Device Architecture (CUDA) C/C++ implementation of the MRT-LBM code. CUDA serves as the computational model for executing computational programs on GPUs, leveraging the inherent advantages of GPU architecture by managing thread and memory hierarchies within the GPU.

The programming model functions by executing computational programs based on the Single Instruction Multiple Thread (SIMT) principle. SIMT involves the parallel execution of multiple threads through a designated function known as the *kernel*, which is enabled by streaming multiprocessors. During the execution of the kernel function,  $N$  distinct kernels operate concurrently in  $N$  distinct grids. Individual threads are identified using the reserved *threadIdx* variable, while thread blocks are accessed through the *blockIdx* variable. The size of the block and grid is determined across one, two, or three dimensions using the variables *blocksize* and *gridsize* respectively [45].

In the current simulation, a performance evaluation was conducted between the GPU-based CUDA C++ implementation for MRT-LBM and CPU-based sequential code, revealing an eighty-fold acceleration compared to CPU-based computations. The CPU configuration comprises an Intel(R) Core(TM) i7-9700K CPU @ 3.60 GHz with 128 GB RAM, while the GPU utilized is a Quadro RTX 6000 with 4608 CUDA cores and 24 GB RAM.



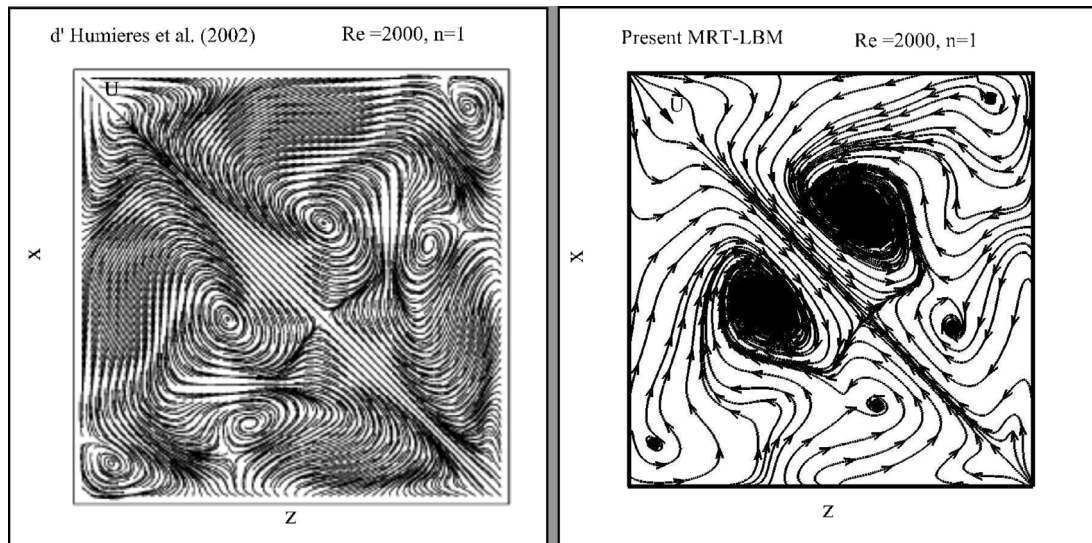


Fig. 3. Comparison of the present MRT-LBM results in terms of the streamlines and velocity vector of d'Humières et al. [46] while  $Re = 2000$ ,  $n = 1$  and  $Bn = 0$ .

#### 4.3. Criteria for convergence

The iterative process in the current MRT-LBM stops when the velocity fields meet the specified convergence criteria.

$$\frac{\sum |\Phi^{l+1} - \Phi^l|}{\sum |\Phi^{l+1}|} < 10^{-9} \quad (23)$$

Here,  $l$  denotes the iteration index,  $\Phi$  represents the velocity field  $\mathbf{u}$ , and the summation is conducted across the entire domain.

### 5. Grid independent test and code validation

To perform a numerical simulation, conducting a grid sensitivity test and validating the code against benchmark solutions is essential. In this simulation, three different lattice sizes ( $192^3$ ,  $224^3$ ,  $256^3$ ) were tested, with the entire simulation conducted using the  $224^3$  lattice dimensions. The subsequent subsections present various code validation tests.

#### 5.1. Code validation for the cubic cavity with top lid-driven diagonally

The initial investigation of a diagonally lid-driven cubic cavity for Newtonian fluid was conducted by d'Humières et al. [46] at a Reynolds number of  $Re = 2000$ . Their study utilized a D3Q15 MRT-LBM with a lattice size of  $64^3$ , employing lid-driven velocity components of  $(u_l, v_l, w_l) = -(\sqrt{2}, 0, \sqrt{2})/20$ . A comparison of the present MRT-LBM results in terms of the streamlines and velocity vector of d'Humières et al. [46] is shown in Fig. 3. This comparison suggests that the current D3Q27 MRT-LBM code is well-suited for simulating diagonally lid-driven cubic cavity flow with Newtonian fluid.

#### 5.2. Code validation for the power-law non-Newtonian fluid in top lid-driven cubic cavity

Fig. 4(a)–(d) depict another code validation test involving a lid-driven cubic cavity with power-law non-Newtonian fluid. The horizontal velocity  $u$  and vertical velocity  $v$  are compared with the results obtained by Adam et al. [47] using the D3Q15 cascaded LBM, considering a Reynolds number of  $Re = 1000$  for both shear-thinning fluid ( $n = 0.8$ ) and shear-thickening fluid ( $n = 1.5$ ). This test confirms the capability of the present MRT-LBM code to simulate power-law non-Newtonian fluid flow characteristics accurately.

#### 5.3. Code validation for the Bingham fluids

The results are compared to previous findings regarding Bingham fluid flow within a square lid-driven chamber to validate the current program, considering  $Bn = 10$  and  $Re = 1000$ . Fig. 5 illustrates the comparison of  $u$ -velocity between the present MRT-LBM study and the work by Syrakos et al. [48] using the finite volume method (FVM) for the Navier–Stokes equations. The lattice size ( $160 \times 160 \times 16$ ) is considered in the present D3Q27 MRT LBM for comparing these results. It is observed that the present code is also suitable for the Bingham fluid.

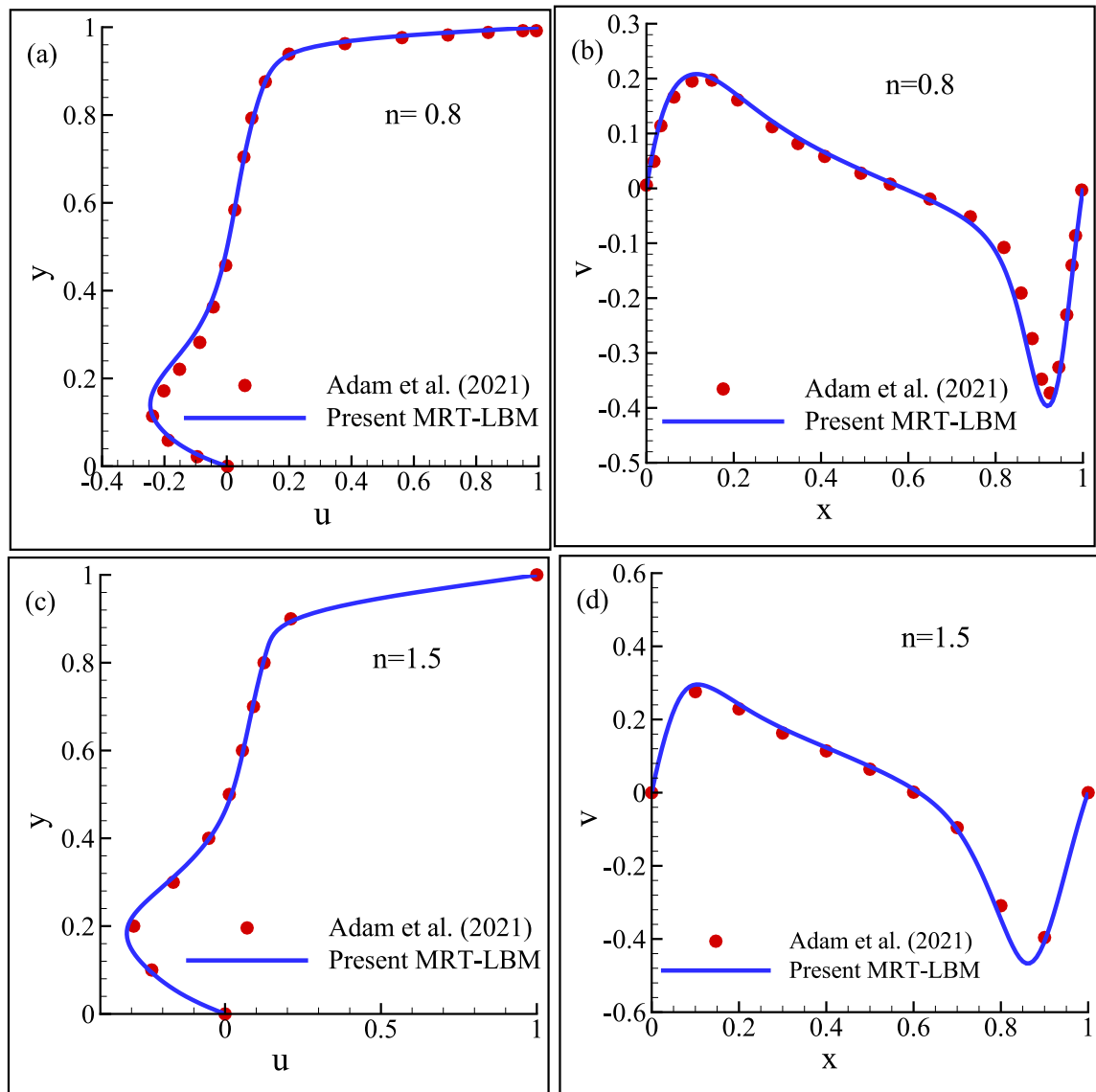


Fig. 4. Contrasting the current results obtained from MRT-LBM for non-Newtonian fluids with the shear-thinning case ( $n = 0.8$ ) using D3Q19 cascaded LBM as presented in Adam et al. [47] (a)  $u$  at  $x = 0.5$  and (b)  $v$  at  $y = 0.5$ . Similarly, for the shear-thickening scenario ( $n = 1.5$ ), the analysis includes: (c)  $u$  at  $x = 0.5$  and (d)  $v$  at  $y = 0.5$ . These comparisons are conducted at a Reynolds number of  $Re = 1000$ , with  $z = 0.5$ .

## 6. Results and discussion

In this current study, Herschel-Bulkley non-Newtonian fluids are employed within the framework of the D3Q27 MRT-lattice Boltzmann method. They are effectively utilized to model 3D laminar flows within a diagonally lid-driven cubic cavity. The governing parameters include  $Re = 300, 600, 1000, 1200$ , and Bingham numbers  $Bn = 0.0, 0.5, 1.0, 2.0$ , with a fixed power-law index of  $n = 0.8$  for the shear-thinning case.

### 6.1. Streamlines and velocity distribution for $Re = 600$

The streamline appended on the  $u$  velocity of 2D flow in the  $xy$ -plane at  $Re = 600$  and  $n = 0.8$  is illustrated in Fig. 6 across various Bingham numbers. At  $Bn = 0$ , the flow pattern exhibits denser regions near the left wall, with small eddies forming in the upper right corner of the cavity. The shear stress distribution is most pronounced near the moving top lid and diminishes towards the bottom wall, influenced by the fluid motion and lid interaction. Since the Bingham number is 0, indicating no yield stress to overcome, stagnant layers do not form near the walls, facilitating uniform fluid flow throughout the cavity. Streamline behavior on



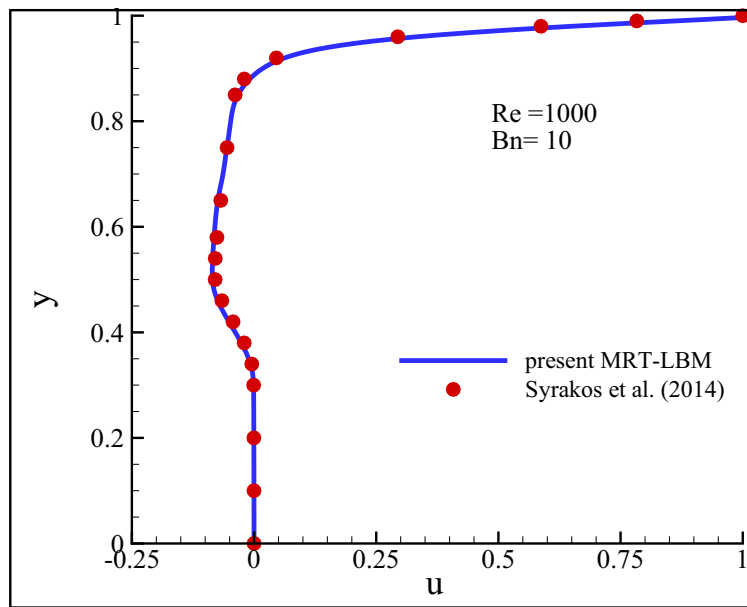


Fig. 5. Comparing the current study with Syrakos et al. [48] about the  $u$ -velocity while  $Re = 1000$ ,  $Bn = 10$  at  $y = 0.5$ .

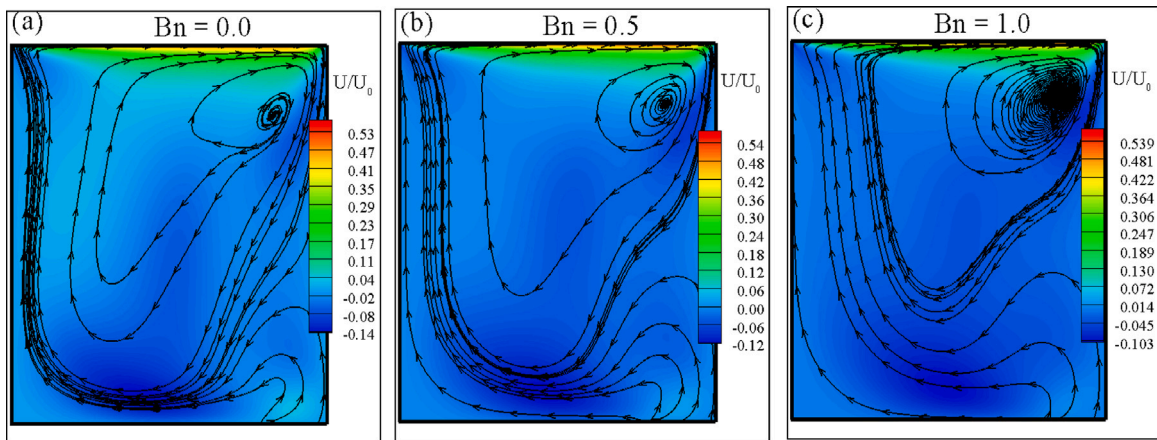


Fig. 6. Streamlines appended on  $u$  velocity for (a)  $Bn = 0.0$  (b)  $Bn = 0.5$  (c)  $Bn = 1$  at  $z/H = 0.5$  while  $Re = 600$  and  $n = 0.8$ .

the  $xy$  plane demonstrates the absence of inactive layers and a smooth, parabolic plug flow profile. The higher Reynolds number induces increased velocities and enhanced mixing, while the power-law index introduces a slight shear-thinning effect near the top lid. Symmetry in the flow pattern is maintained due to boundary conditions and geometric symmetry. With an increase in Bingham number depicted in Fig. 6(b) and (c), the size of eddies enlarges, particularly noticeable at  $Bn = 1.0$ . The progression from a Bingham number of 0 to 0.5 and 1 notably influences the streamline behavior in the  $xy$  plane of the top lid-driven cavity. This altered flow regime is characterized by the emergence of stagnant layers, reduction in the moving core, highly non-uniform flow, and loss of symmetry. Comprehending these variations requires thoroughly understanding Bingham plastic rheology and its interactions with shear stress and non-Newtonian effects.

Fig. 7 illustrates streamlines at the  $zx$ -plane for different Bingham numbers ( $Bn = 0, 0.5, 1$ ) with a Reynolds number of  $Re = 600$  and a power-law index of  $n = 0.8$ . In Fig. 7(a) ( $Bn = 0.0$ ), the maximum fluid flow is observed, generating two primary eddies along the middle diagonally intersecting sides. However, these middle circulations diminish for  $Bn = 0.5$  in Fig. 7(b). Here, the fluid flows diagonally on both sides of the intersection, converging at the left bottom side of the cavity. Similarly, in Fig. 7(c), as the Bingham number increases to  $Bn = 1$ , the fluid flow pattern resembles that of Fig. 7(b), but with the addition of two eddies created at the top-left and bottom-right sides of the cavity. The Bingham number signifies the ratio of yield stress to viscous stress. A Newtonian fluid possesses a  $Bn$  value of 0, while Bingham plastic behavior emerges with higher  $Bn$  values. The contrast between two Bingham plastic fluids with distinct yield stresses ( $Bn = 0.5$  and 1) and a Newtonian fluid ( $Bn = 0$ ) is evident. Additionally, the power-law

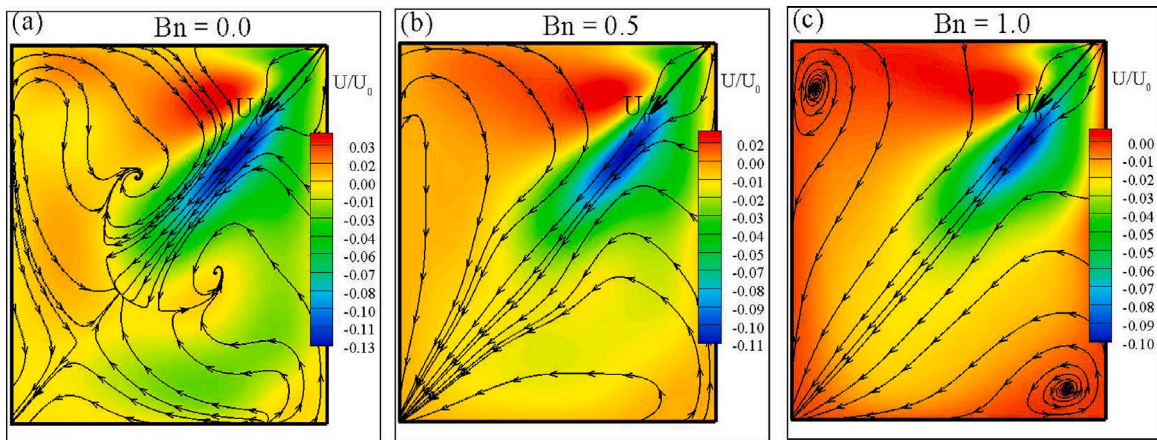


Fig. 7. Streamlines appended on  $u$  velocity for (a)  $Bn = 0.0$  (b)  $Bn = 0.5$  (c)  $Bn = 1$  at  $y/H = 0.5$  while  $Re = 600$  and  $n = 0.8$ .

index characterizes the shear-thinning behavior of the fluid, where a value of  $n = 0.8$  indicates shear-thinning, resulting in decreasing viscosity as the shear rate increases. This can lead to more complex flow patterns than Newtonian fluids. Increasing  $Bn$  tends to suppress flow near the walls due to the elevated yield stress, thickening the boundary layer and decreasing flow velocity near the walls. Consequently, this may lead to the formation of recirculation zones in the cavity's corners. At higher Reynolds numbers ( $Re$ ), inertial forces become more prominent, potentially inducing flow instabilities, vortices, and other irregularities.

The impact of the Bingham number on velocity distribution within the  $zx$  plane is depicted in Fig. 8, maintaining a Reynolds number of  $Re = 600$  and a power-law index of  $n = 0.8$ . In Fig. 8(a), the  $u$ -velocity exhibits an increase corresponding to the rising Bingham number. Within the range  $0 \leq y \leq 0.08$ , the  $u$ -velocity escalates by 50% as the Bingham number transitions from 0 to 2. Near the upper wall at approximately  $y = 1$ , the  $u$ -velocity remains nearly constant. Conversely, in Fig. 8(b), the  $v$ -velocity diminishes as the Bingham number increases within the domain  $0 \leq x \leq 0.25$  near the left wall. However, this trend shifts within the range  $0.25 \leq x \leq 1$ , where the velocity ascends with the elevation of the Bingham number. At low values of  $Bn$ , the yield stress characteristic of Bingham plastic fluids becomes pertinent, forming a thin, motionless layer near the walls. This layer impedes velocity and constrains flow within the core region. As  $Bn$  increases, the yield stress gains prominence, further thickening the stationary layer. However, this phenomenon propels the remaining fluid towards the center of the cavity, resulting in a more focused flow with increased velocity. The continuous motion of the top lid generates shear stress, propelling the flow within the cavity. Even in the presence of the stationary layer induced by yield stress, this shear stress surpasses the yield stress barrier, compelling the fluid to move. As the Bingham number escalates, so does the yield stress, demanding a greater shear stress threshold for overcoming it. Nonetheless, the perpetual lid motion ensures the presence of the requisite shear stress, thus fostering higher velocity within the core region to counterbalance the thicker stationary layer.

## 6.2. Streamlines and velocity distribution for $Re = 1200$

At a higher Reynolds number ( $Re = 1200$ ), Fig. 9 illustrates streamlines for various Bingham numbers with  $n = 0.8$  at the  $xy$ -plane ( $z/H = 0.5$ ). The fluid flow behavior undergoes modifications, with the Bingham number increasing from 0 to 1 at this elevated Reynolds number. At  $Bn = 0$  in Fig. 9(a), a circulation pattern forms in the middle of the cavity. This circulation moves towards the upper right position as the Bingham number increases to 0.5, as depicted in Fig. 9(b). In Fig. 9(c) at  $Bn = 1$ , the circulation remains consistent with its previous state, but the fluid flow intensifies. The outcomes for  $Bn = 0.5$  and  $Bn = 1.0$  closely resemble those observed at  $Re = 600$ . In this context,  $Re = 1200$  denotes thermal conditions that may induce buoyancy-driven flows within the cavity. Temperature disparities could lead to interactions with the lid-driven flow, influencing the overall circulation pattern. Buoyancy effects might complement or counteract the circulation towards the upper right side, contingent upon temperature distribution and its interplay with yield and shear stress non-uniformities. The constant motion of the top lid generates shear stress, propelling the flow within the cavity. However, this non-uniformity can result in asymmetric flow patterns, with fluid preferentially moving towards higher-stress regions. The elevated stress region might be situated on the upper right side of the  $xy$ -plane, influencing the direction of circulation, depending on the specific configuration. Shear thinning, reducing viscosity in regions of higher shear stress, such as the potential non-uniformity area on the upper right side, facilitates freer fluid flow, contributing to the observed circulation pattern.

Fig. 10(a)–(c) depict streamlines at the  $zx$ -plane for different Bingham numbers ( $Bn = 0, 0.5, 1$ ) at a Reynolds number of  $Re = 1200$ . In Fig. 10(a), reminiscent of the steady-state flow pattern observed at Bingham number,  $Bn = 0$ , four distinct vortex groups can be identified. The base flow vorticity field appears strained and antisymmetric near the diagonal interface plane. As depicted in Fig. 10(b), the eddies disintegrate as the Bingham number increases from 0 to 0.5. In this scenario, the flow from both sides converges in the middle of the intersection. However, when the Bingham number reaches 1, as shown in Fig. 10(c), the fluid

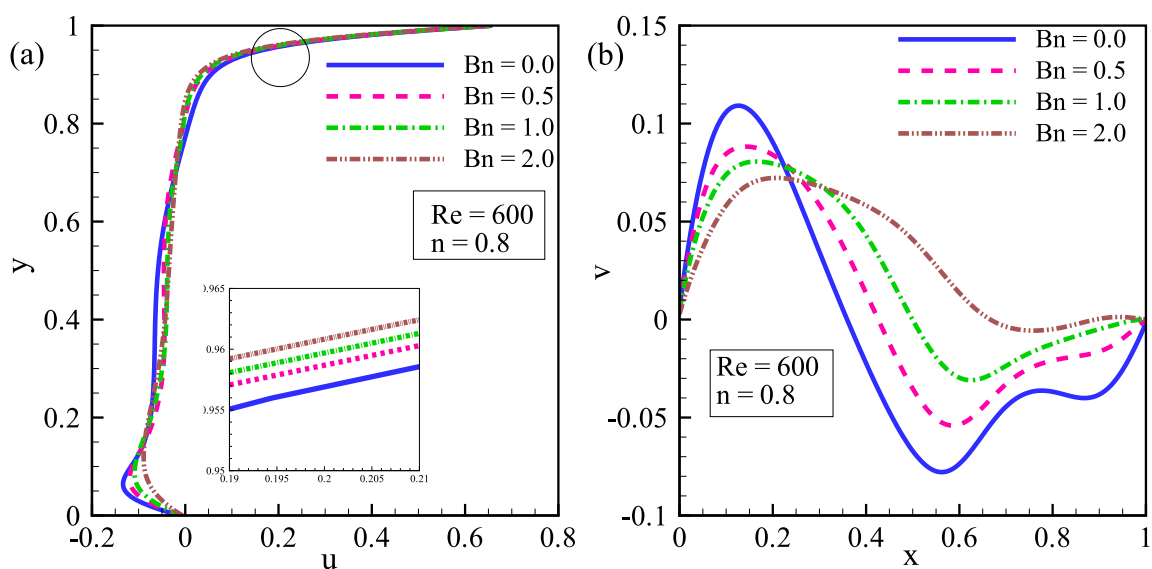


Fig. 8. Velocity distribution for the different Bingham number ( $Bn = 0.0, 0.5, 1.0, 2.0$ ) (a)  $u/U$  at  $x/H = 0.5$  and (b)  $v/U$  at  $y/H = 0.5$  while  $Re = 600$  and  $z/H = 0.5$ .

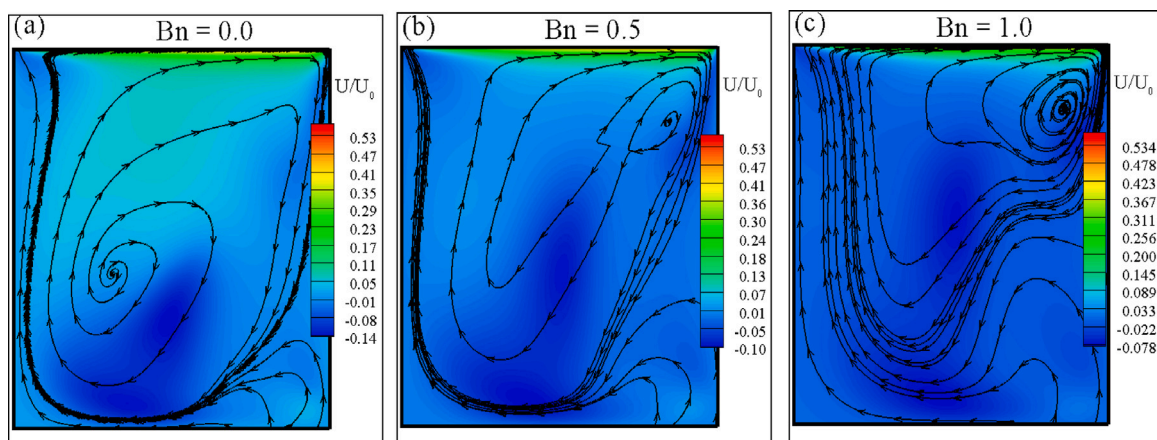


Fig. 9. Streamlines appended on  $u$  velocity for (a)  $Bn = 0.0$  (b)  $Bn = 0.5$  (c)  $Bn = 1$  at  $z/H = 0.5$  while  $Re = 1200$  and  $n = 0.8$ .

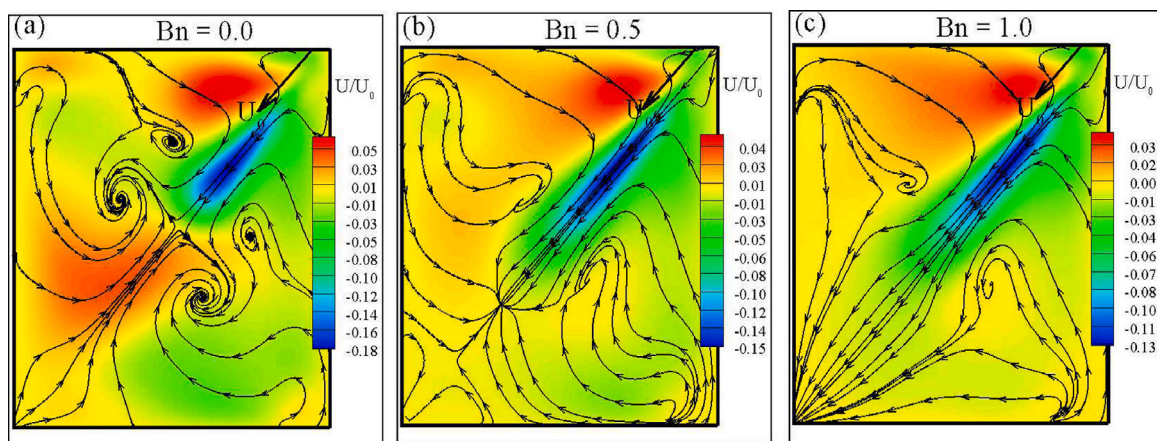


Fig. 10. Streamlines appended on  $u$  velocity for (a)  $Bn = 0.0$  (b)  $Bn = 0.5$  (c)  $Bn = 1$  at  $y/H = 0.5$  while  $Re = 1200$  and  $n = 0.8$ .

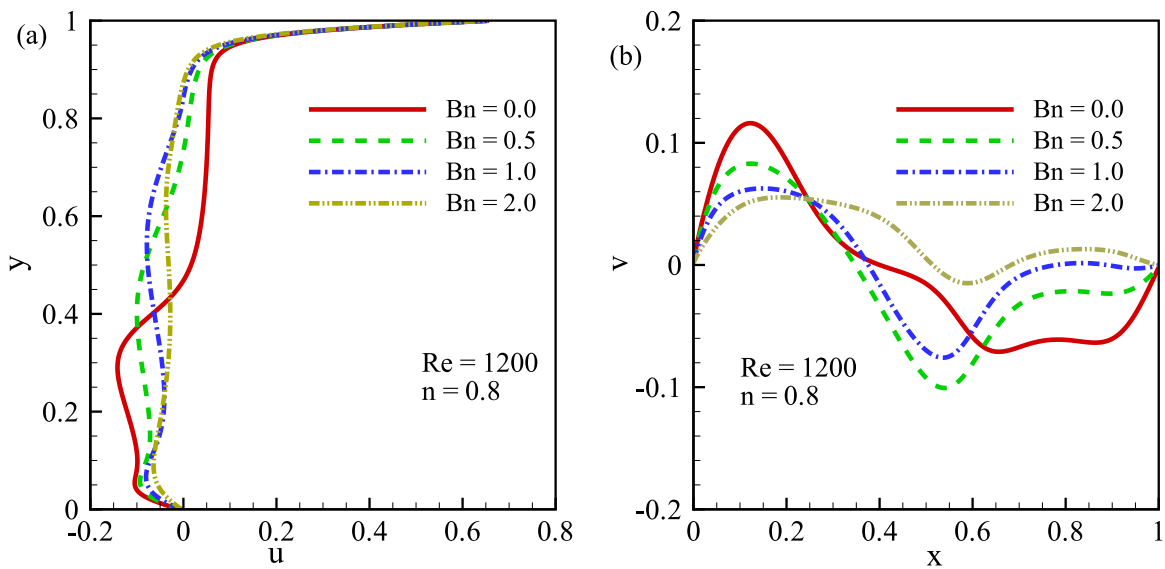


Fig. 11. Velocity distribution for the different Bingham number ( $Bn = 0.0, 0.5, 1.0, 2.0$ ) (a)  $u/U$  at  $x/H = 0.5$  and (b)  $v/U$  at  $y/H = 0.5$  while  $Re = 1200$  and  $z/H = 0.5$ .

flow aligns in identical directions on both sides of the intersection, streaming towards the left-bottom corner. The high Reynolds number and complex flow interactions may form secondary vortices within the moving core, intensifying mixing and contributing to flow irregularity. The discharge may exhibit turbulent characteristics at a high Reynolds number of 1200, particularly near the jet-like regions and the secondary vortices. This turbulence could further obscure flow behavior and heighten the accuracy prediction challenge. Key features of this flow regime include a small moving core, significant concentration of shear stress, formation of secondary vortices, limited mixing, prominent stationary layers, highly non-uniform flow, and the potential for turbulent flow.

In the preceding velocity distributions displayed in Fig. 8, the Reynolds number was 600. However, the Reynolds number increased to 1200 in Fig. 11, while the other parameters remained unchanged. Once again, the Bingham number varies across this figure. In Fig. 11(a), the  $u$ -velocity increases as the Bingham number rises within the range  $0 \leq y \leq 0.4$ . The  $v$ -velocity, the other hand, decreases by 33.33% and 50% when  $Bn$  increases from 0 to 0.5 and 0 to 1.0, respectively. Subsequently, as the Bingham number increases to 2, the  $v$ -velocity decreases by 58.33%, after which the velocity declines. The maximum velocity is observed for  $Bn = 0$  within the  $0.4 \leq y \leq 0.8$  range. The  $u$ -velocity remains nearly constant near the upper wall. Conversely, the velocity decreases as the Bingham number rises within the domain  $0 \leq x \leq 0.25$ , as depicted in Fig. 11(b). Within the  $0.25 \leq x \leq 0.6$  range, the  $v$ -velocity decreases as the Bingham number increases from 0 to 1 but rises when  $Bn = 2$ . In the remaining scenario,  $0.6 \leq x \leq 1$ , the rate increases again with the rising Bingham number. Tabulated profiles of  $u$ -velocity along the  $y$ -coordinates and  $v$ -velocity along the  $x$ -coordinates are presented in Tables 1 and 2 respectively, for Reynolds numbers of 1000 and 1200, and Bingham numbers of 0.0, 0.5, 2.0, with  $n = 0.8$ . The  $u$ -velocity along  $y$ -coordinates increases 234.11% and 1.91% for  $y = 0.5$  and  $y = 1.0$ , respectively, at  $Bn = 0.0$  when the Reynolds number increases from 100 to 1200. On the other hand, the velocity profile along the  $x$ -coordinate reduces 5.04% and 2.08% for  $Bn = 0.0$  and  $Bn = 2$  respectively and rises 5.5% for  $Bn = 0.5$  at  $x = 0.05$  when the Reynolds number increases from 1000 to 1200. The findings in Fig. 8 (for  $Re = 600$ ) closely resemble those of Fig. 11 (for  $Re = 1200$ ).

### 6.3. Vorticity distribution

The vorticity distribution of the base flow, evaluated at the mid-span cross-section of the cavity, is depicted in Fig. 12(a)–(c). Represented in the  $zx$  plane, it is computed as the curl of the velocity field averaged over a complete period. The contours are overlaid with representative base flow pathlines projected onto the mid-span plane, resembling a topographic flow map. Closed shapes in this map indicate the location and size of corresponding vortex cores. Fig. 12 showcases the vorticity distribution in the  $zx$  plane for three distinct Reynolds numbers ( $Re = 300, 600$ , and 1200) within a cubic cavity under top lid-driven conditions, with a fixed Bingham number of 0.5 and a power-law index of 0.8. These figures provide insights into the local rotation of the fluid, with stronger rotation associated with higher vorticity values. The color-coded outlines denote vorticity, where blue represents negative vorticity (clockwise rotation), red signifies positive vorticity (counterclockwise rotation), and white patches denote zero vorticity. As depicted in Fig. 12(a) for  $Re = 300$ , most of the vorticity is concentrated in the boundary layers near the cavity's bottom and sides, with two weaker vortices in the upper corners. Fig. 12(b) for  $Re = 600$  exhibits four distinct vortices, one in each corner, with vorticity more evenly distributed throughout the cavity. Additionally, these vortices display greater resilience than those observed at  $Re = 300$ . In Fig. 12(c) for  $Re = 1200$ , the vorticity distribution becomes significantly more complex and turbulent. The cavity hosts



**Table 1**Tabulated  $u$ -velocity profiles along the  $y$ -coordinate of the lid-driven cavity for various values of  $Re = 1000, 1200$  and  $Bn = 0.0, 0.5, 2.0$ .

| y-coordinate | $Re = 1000$ |            |            | $Re = 1200$ |            |            |
|--------------|-------------|------------|------------|-------------|------------|------------|
|              | $Bn = 0.0$  | $Bn = 0.5$ | $Bn = 2.0$ | $Bn = 0.0$  | $Bn = 0.5$ | $Bn = 2.0$ |
| 0.0          | −0.0112     | −0.0096    | −0.0053    | −0.0079     | −0.0068    | −0.0035    |
| 0.05         | −0.1113     | −0.0980    | −0.0630    | −0.1047     | −0.0930    | −0.0571    |
| 0.10         | −0.0996     | −0.0839    | −0.0734    | −0.0989     | −0.0783    | −0.0635    |
| 0.15         | −0.0997     | −0.0654    | −0.0658    | −0.1072     | −0.0716    | −0.0556    |
| 0.20         | −0.1075     | −0.0605    | −0.0552    | −0.1217     | 0.0751     | −0.0461    |
| 0.25         | −0.1157     | −0.0618    | −0.0460    | −0.1362     | −0.0822    | −0.0384    |
| 0.30         | −0.1188     | −0.0660    | −0.0387    | −0.1412     | −0.0902    | −0.0327    |
| 0.35         | −0.1103     | −0.0710    | −0.0338    | −0.1204     | −0.0969    | −0.0291    |
| 0.40         | −0.0852     | −0.0752    | −0.0309    | −0.0623     | −0.0999    | −0.0282    |
| 0.45         | −0.0476     | −0.0771    | −0.0298    | −0.0109     | −0.0958    | −0.0294    |
| 0.50         | −0.0129     | −0.0752    | −0.0295    | 0.0173      | −0.0836    | −0.0320    |
| 0.55         | 0.0103      | −0.0684    | −0.0292    | 0.0317      | −0.0647    | −0.0351    |
| 0.60         | 0.0244      | −0.0567    | −0.0278    | 0.0402      | −0.0411    | −0.0372    |
| 0.65         | 0.0333      | −0.0414    | −0.0247    | 0.0457      | −0.0234    | −0.0367    |
| 0.70         | 0.0392      | −0.0248    | −0.0201    | 0.0497      | −0.0069    | −0.0317    |
| 0.75         | 0.0431      | −0.0109    | −0.0157    | 0.0521      | 0.0041     | −0.0238    |
| 0.80         | 0.0472      | 0.0037     | −0.0093    | 0.0537      | 0.0119     | −0.0153    |
| 0.85         | 0.0508      | 0.0130     | −0.0022    | 0.0555      | 0.0199     | −0.0057    |
| 0.90         | 0.0603      | 0.0354     | 0.0144     | 0.0616      | 0.0346     | 0.0110     |
| 0.95         | 0.1191      | 0.1070     | 0.0869     | 0.1101      | 0.1004     | 0.0777     |
| 1.00         | 0.6446      | 0.6446     | 0.6420     | 0.6569      | 0.6571     | 0.6549     |

numerous vortices of varying sizes and intensities, and the flow exhibits erratic behavior, causing continuous shifts in the orientation and formation of the vortices. This graphic illustrates the influence of the Reynolds number on vorticity distribution within a cubic cavity under top lid-driven flow. As the Reynolds number increases, the flow becomes more chaotic, and the vorticity distribution becomes increasingly intricate.

#### 6.4. Iso-surface of non-Newtonian viscosity

The iso-surface representation of viscosity distribution provides insights into the spatial variation of viscosity within the fluid domain under different Reynolds number ( $Re$ ) and Bingham number ( $Bn$ ) conditions. At  $Re = 1200$ , four distinct cases of Bingham number ( $Bn$ ) are examined: (a)  $Bn = 0.2$ , (b)  $Bn = 0.5$ , (c)  $Bn = 1$ , and (d)  $Bn = 2$ . Each iso-surface visually delineates regions within the fluid domain where viscosity reaches a specific value, facilitating the identification of areas with varying viscosity levels. This is crucial for comprehending the rheological behavior of the fluid under investigation. Examining these iso-surfaces for different combinations of  $Re$  and  $Bn$  yields insights into how these parameters influence the distribution and magnitude of viscosity within the fluid. Such visualization aids in analyzing and interpreting fluid flow characteristics, enhancing understanding of complex fluid systems' dynamics. In Fig. 13, a three-dimensional representation shows a sphere with a central void filled with fluid, with varying Bingham numbers across different sections. The Bingham number quantifies a fluid's non-Newtonian behavior, representing its yield stress relative to its plastic viscosity. The iso-surfaces in the four sections of the figure represent locations in space where viscosity remains constant. The color of these iso-surfaces indicates the viscosity value. For instance, at  $Bn = 0.2$  in Fig. 13(a), the fluid exhibits high plastic viscosity and low yield stress, resulting in a relatively spherical and smooth iso-surface. Moving to Fig. 13(b) with  $Bn = 0.5$ , the fluid's plastic viscosity remains relatively low, but its yield stress is higher, leading to a somewhat distorted iso-surface compared to Fig. 13(a). Fig. 13(c) demonstrates a fluid with low plastic viscosity and high yield stress ( $Bn = 1.0$ ), resulting in a highly deformed iso-surface with sharp edges. Finally, in Fig. 13(d) where  $Bn = 2.0$ , the fluid's plastic viscosity remains very low while its yield stress is high, causing significant distortion of the iso-surface, almost resembling a cube. This visualization highlights the considerable influence of the Bingham number on the shape of the viscosity distribution iso-surface. As the Bingham number increases, the iso-surface exhibits sharper edges and more significant distortion, reflecting the nonlinear flow behavior of the fluid due to its yield stress.

#### 6.5. Comparison between the D3Q19 and D3Q27 MRT-LBM

The detailed comparison of D3Q19 and D3Q27 MRT-LBM (Multiple-Relaxation-Time Lattice Boltzmann Methods) models reveals that both discretizations provide excellent agreement in predicting the velocity distributions for various Bingham numbers at a Reynolds number  $Re = 1200$ , which are shown in Fig. 14(a)–(b). The details of the D3Q19 MRT-LBM were given in [42]. The comparison between D3Q19 and D3Q27 MRT-LBM models focuses on how these models predict the velocity distribution under different conditions, specifically for various Bingham numbers ( $Bn = 0.0, 0.2, 0.5$ ). The velocity components are evaluated at specific points in the domain. The D3Q27 model, with more discrete velocity directions, might offer slight improvements in capturing complex flow behaviors, especially for higher Reynolds numbers. However, both models are sufficiently accurate for practical purposes, demonstrating their robustness and reliability in simulating Newtonian and non-Newtonian fluid flows.

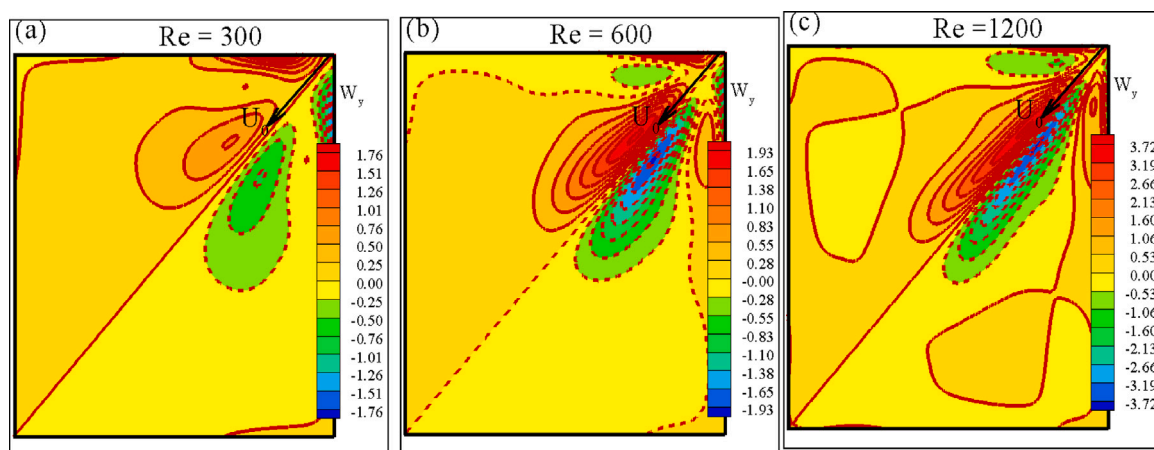


Fig. 12. Vorticity distribution ( $\omega_\eta = \frac{\partial w}{\partial \xi} - \frac{\partial u}{\partial \zeta}$ ) at  $z$ - $x$ -plane for (a)  $Re = 300$  (b)  $Re = 600$  (c)  $Re = 1200$ , while  $n = 0.8$ ,  $Bn = 0.5$  at  $y/H = 0.5$ .

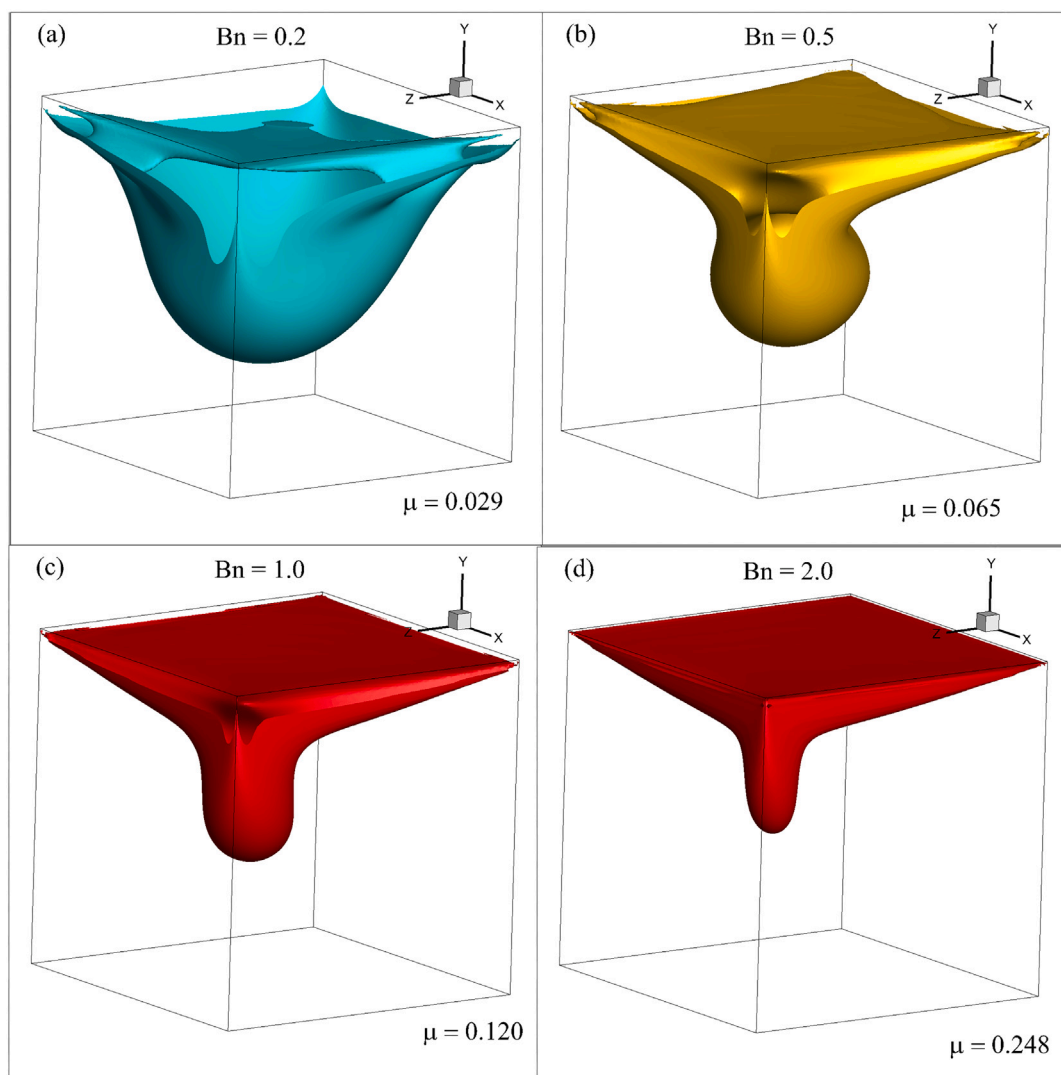


Fig. 13. Iso-surface of viscosity distribution for  $Re = 1200$  (a)  $Bn = 0.2$  (b)  $Bn = 0.5$  (c)  $Bn = 1$  (d)  $Bn = 2$ .



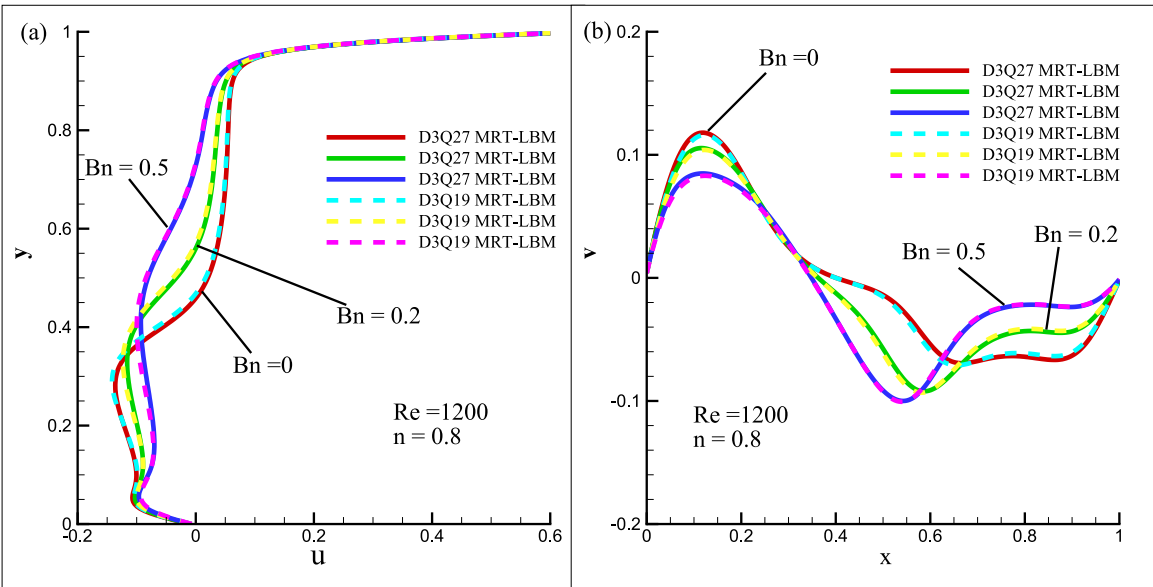


Fig. 14. Comparison between the D3Q19 and D3Q27 MRT-LBM in terms of the velocity distribution for the different Bingham number ( $Bn = 0.0, 0.2, 0.5$ ) (a)  $u/U$  at  $x/H = 0.5$  and (b)  $v/U$  at  $y/H = 0.5$  while  $Re = 1200$  and  $z/H = 0.5$ .

Table 2

Tabulated  $v$ -velocity profiles along the  $x$ -coordinate of the lid-driven cavity for various values of  $Bn = 0.0, 0.5, 2.0$  and  $Re = 1000, 1200$ .

| x-coordinate | $Re = 1000$ |            |          | $Re = 1200$ |            |          |
|--------------|-------------|------------|----------|-------------|------------|----------|
|              | $Bn = 0$    | $Bn = 0.5$ | $Bn = 2$ | $Bn = 0$    | $Bn = 0.5$ | $Bn = 2$ |
| 0.0          | 0.0059      | 0.0051     | 0.0026   | 0.0040      | 0.0038     | 0.0018   |
| 0.05         | 0.0814      | 0.0617     | 0.0337   | 0.0773      | 0.0651     | 0.0330   |
| 0.10         | 0.1139      | 0.0788     | 0.0517   | 0.1137      | 0.0818     | 0.0485   |
| 0.15         | 0.1115      | 0.0792     | 0.0591   | 0.1107      | 0.0810     | 0.0545   |
| 0.20         | 0.0875      | 0.0711     | 0.0602   | 0.0841      | 0.0703     | 0.0552   |
| 0.25         | 0.0552      | 0.0554     | 0.0586   | 0.0505      | 0.0519     | 0.0537   |
| 0.30         | 0.0269      | 0.0323     | 0.0558   | 0.0248      | 0.0272     | 0.0509   |
| 0.35         | 0.0068      | 0.0036     | 0.0520   | 0.0091      | −0.0014    | 0.0460   |
| 0.40         | −0.0074     | −0.0279    | 0.0461   | −0.0004     | −0.0357    | 0.0380   |
| 0.45         | −0.0211     | −0.0577    | 0.0360   | −0.0067     | −0.0677    | 0.0237   |
| 0.50         | −0.0415     | −0.0795    | 0.0204   | −0.0160     | −0.0932    | 0.0045   |
| 0.55         | −0.0683     | −0.0857    | 0.0036   | −0.0348     | −0.0998    | −0.0109  |
| 0.60         | −0.0835     | −0.0693    | −0.0064  | −0.0595     | −0.0801    | −0.0146  |
| 0.65         | −0.0780     | −0.0435    | −0.0065  | 0.0709      | −0.0514    | −0.0069  |
| 0.70         | −0.0652     | −0.0267    | −0.0010  | −0.0667     | −0.0314    | 0.0039   |
| 0.75         | −0.0573     | −0.0193    | 0.0041   | −0.0619     | −0.0238    | 0.0099   |
| 0.80         | −0.0562     | −0.0171    | 0.0069   | −0.0614     | −0.0215    | 0.0125   |
| 0.85         | −0.0586     | −0.0177    | 0.0079   | −0.0634     | −0.0223    | 0.0130   |
| 0.90         | −0.0555     | −0.0190    | 0.0069   | −0.0601     | −0.0234    | 0.0111   |
| 0.95         | −0.0382     | −0.0150    | 0.0035   | −0.0394     | −0.0181    | 0.0056   |
| 1.00         | −0.0026     | −0.0012    | 0.0001   | −0.0019     | −0.0011    | 0.0001   |

Table 3

Simulation time (s) for the D3Q19 and D3Q27 MRT-LBM while  $Re = 1200$  and  $Bn = 0.5$  for the different grid with 10000 steps iteration. GPU: Quadro RTX 6000 with 4608 CUDA cores and 24 GB RAM.

| Lattice Size |         |         |         |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|---------|---------|---------|
| MRT-LBM      | $160^3$ | $192^3$ | $224^3$ | $256^3$ | $288^3$ | $320^3$ | $352^3$ | $384^3$ |
| D3Q19        | 117.71  | 180.37  | 292.77  | 404.61  | 686.39  | 944.34  | 1152.92 | 1489.37 |
| D3Q27        | 231.64  | 438.39  | 626.11  | 981.21  | 1422.08 | 1916.12 | 2356.94 | 3305.92 |

The multi-relaxation-time lattice Boltzmann method is a powerful technique for simulating fluid flow in computational fluid dynamics. Among the different lattice configurations, D3Q19 and D3Q27 are two commonly used three-dimensional models, where the numbers represent the discrete velocity directions used in the simulation. Simulation time is a critical factor in evaluating the efficiency of these models. A comparative analysis of the D3Q19 and D3Q27 MRT-LBM reveals a significant difference in their

computational demands, shown in Table 3 for the different grid arrangements. Specifically, the D3Q27 MRT-LBM requires more than double the simulation time compared to the D3Q19 MRT-LBM. The D3Q27 model utilizes 27 discrete velocity directions, whereas the D3Q19 model uses only 19. This increase in the number of directions leads to a corresponding rise in distribution functions that must be computed and stored at each lattice node. In the MRT-LBM, the collision operator transforms the distribution functions into moment space, performs relaxation, and then transforms them back. The complexity of these operations scales with the number of velocity directions. Consequently, the D3Q27 model involves more complex and time-consuming calculations for each time step. The D3Q27 model requires more memory to store additional distribution functions, leading to increased memory bandwidth usage and potential bottlenecks in data transfer within the computational architecture. More velocity directions mean more calculations per lattice node. This results in a higher computational load per time step, leading to longer overall simulation times. Empirical studies and simulations confirm these theoretical expectations. For instance, in a benchmark simulation involving fluid flow in a three-dimensional cavity, it was observed that the D3Q27 MRT-LBM took more than twice the simulation time compared to the D3Q19 MRT-LBM for the same physical scenario and resolution. In summary, while the D3Q27 MRT-LBM offers potentially higher accuracy due to its finer discretization of velocity space, this comes at the cost of significantly increased simulation time. Therefore, the choice between the D3Q19 and D3Q27 models involves a trade-off between computational efficiency and accuracy, with the D3Q27 MRT-LBM requiring more than double the time compared to the D3Q19 MRT-LBM for typical simulations.

## 7. Conclusions

This paper investigates the GPU-accelerated D3Q27 Multiple-Relaxation-Time Lattice Boltzmann Simulation of Herschel-Bulkley non-Newtonian Fluids within a Cubic Cavity driven diagonally by a top lid. The study explores variations in parameters such as Reynolds number ( $Re = 300, 600, 1000, 1200$ ), Bingham number ( $Bn = 0.0, 0.5, 1.0, 2.0$ ), and Power-law index ( $n = 0.8$ ), alongside other fixed parameters. Computational solutions are visually represented to provide both numerical and physical interpretations of the results. The validity of the current MRT-LBM formulation is confirmed through multiple validations against previous research works. Key findings from the analysis are summarized as follows:

- As  $Bn$  increases, stagnant layers emerge near the walls, disrupting the flow and reducing core movement.
- Higher  $Bn$  values at  $Re = 600$  cause larger eddies and non-uniform flow, especially in the  $xy$  plane, altering streamline behavior.
- Increasing  $Re$  to 1200 shows more complex flow patterns, improved circulation, and vortex formation with potential intensity.
- Higher  $Bn$  suppresses flow near walls and increases core shear stress, significantly affecting velocity distribution and flow characteristics.
- $Re = 300$  concentrates most of the vorticity in the boundary layers along the cavity's bottom and sides, with two weaker vortices in the top corners.  $Re = 600$  has four distinct vortices, one in each corner, with vorticity more equally distributed throughout the cavity.
- As the Reynolds number increases, the flow becomes more chaotic, and the vorticity distribution becomes increasingly intricate.
- Iso-surface representations of viscosity highlight the spatial variability of viscosity, showing increased distortion with rising  $Bn$ , illustrating the non-linear behavior of the fluid due to yield stress effects.
- The numerical outcomes obtained in this study are poised to serve as benchmark results for validating 3D codes concerning non-Newtonian power-law and Herschel-Bulkley fluids.
- Simulation times for the D3Q19 and D3Q27 MRT-LBM models indicate that the D3Q27 MRT-LBM requires more than twice the time of the D3Q19 MRT-LBM.

## CRedit authorship contribution statement

**Md. Mamun Molla:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Md. Mahadul Islam:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix

The obtained  $27 \times 27$  orthogonal transformation matrix  $\mathbf{M}$  for the D3Q27 lattice model is give below:

[illegible]

The inverse of this **M** matrix is used in this study which is given below:

|                                   |   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |      |    |     |     |     |     |     |     |     |     |   |   |   |   |   |   |   |   |   |   |   |
|-----------------------------------|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|----|-----|-----|-----|-----|-----|-----|-----|-----|---|---|---|---|---|---|---|---|---|---|---|
| $\mathbf{M}^{-1} = \frac{1}{216}$ | 8 | 0   | 0   | 0   | -24 | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 24  | -108 | 0  | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |   |
|                                   | 8 | 12  | 0   | 0   | -12 | 12  | 0   | 0   | 0   | 0   | -12 | 0   | 0   | 12  | 0   | 0   | 0    | 4  | -12 | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |   |
|                                   | 8 | 0   | 12  | 0   | -12 | -6  | 18  | 0   | 0   | 0   | 0   | -12 | 0   | 0   | 12  | 0   | 0    | 4  | 6   | -18 | 0   | 0   | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |   |
|                                   | 8 | -12 | 0   | 0   | -12 | 12  | 0   | 0   | 0   | 0   | 12  | 0   | 0   | -12 | 0   | 0   | 0    | 4  | -12 | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |   |
|                                   | 8 | 0   | -12 | 0   | -12 | -6  | 18  | 0   | 0   | 0   | 0   | 12  | 0   | 0   | -12 | 0   | 0    | 4  | 6   | -18 | 0   | 0   | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |   |
|                                   | 8 | 0   | 0   | 12  | -12 | -6  | -18 | 0   | 0   | 0   | 0   | 0   | -12 | 0   | 0   | 12  | 0    | 4  | 6   | 18  | 0   | 0   | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |   |
|                                   | 8 | 0   | 0   | -12 | -12 | -6  | -18 | 0   | 0   | 0   | 0   | 0   | 12  | 0   | 0   | -12 | 0    | 4  | 6   | 18  | 0   | 0   | 0   | 0   | 0   | 0   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |   |
|                                   | 8 | 12  | 12  | 0   | 0   | 6   | 18  | 18  | 0   | 0   | -3  | -3  | 0   | -6  | -6  | 0   | -6   | -2 | 3   | 9   | -18 | 0   | 0   | 0   | 27  | -27 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |   |
|                                   | 8 | -12 | 12  | 0   | 0   | 6   | 18  | -18 | 0   | 0   | 3   | -3  | 0   | 6   | -6  | 0   | -6   | -2 | 3   | 9   | 18  | 0   | 0   | -27 | -27 | 0   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |   |
|                                   | 8 | -12 | -12 | 0   | 0   | 6   | 18  | 18  | 0   | 0   | 3   | 3   | 0   | 6   | 6   | 0   | -6   | -2 | 3   | 9   | -18 | 0   | 0   | -27 | 27  | 0   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|                                   | 8 | 12  | -12 | 0   | 0   | 6   | 18  | -18 | 0   | 0   | -3  | 3   | 0   | -6  | 6   | 0   | -6   | -2 | 3   | 9   | 18  | 0   | 0   | 27  | 27  | 0   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|                                   | 8 | 12  | 0   | 12  | 0   | 6   | -18 | 0   | 0   | 18  | -3  | 0   | -3  | -6  | 0   | -6  | -6   | -2 | 3   | -9  | 0   | 0   | -18 | -27 | 0   | 27  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|                                   | 8 | 0   | 12  | 12  | 0   | -12 | 0   | 0   | 18  | 0   | 0   | -3  | -3  | 0   | -6  | -6  | -6   | -2 | -6  | 0   | 0   | -18 | 0   | 0   | 27  | -27 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|                                   | 8 | -12 | 0   | 12  | 0   | 6   | -18 | 0   | 0   | -18 | 3   | 0   | -3  | 6   | 0   | -6  | -6   | -2 | 3   | -9  | 0   | 0   | 18  | 27  | 0   | 27  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|                                   | 8 | 0   | -12 | 12  | 0   | -12 | 0   | 0   | -18 | 0   | 0   | 3   | -3  | 0   | 6   | -6  | -6   | -2 | -6  | 0   | 0   | 18  | 0   | 0   | -27 | -27 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|                                   | 8 | 12  | 0   | -12 | 0   | 6   | -18 | 0   | 0   | -18 | -3  | 0   | 3   | -6  | 0   | 6   | -6   | -2 | 3   | -9  | 0   | 0   | 18  | -27 | 0   | -27 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|                                   | 8 | 0   | 12  | -12 | 0   | -12 | 0   | 0   | -18 | 0   | 0   | -3  | 3   | 0   | -6  | 6   | -6   | -2 | -6  | 0   | 0   | 18  | 0   | 0   | 27  | 27  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|                                   | 8 | -12 | 0   | -12 | 0   | 6   | -18 | 0   | 0   | 18  | 3   | 0   | 3   | 6   | 0   | 6   | -6   | -2 | 3   | -9  | 0   | 0   | -18 | 27  | 0   | -27 |   |   |   |   |   |   |   |   |   |   |   |

(24)

## Data availability

This study is not associated with any third party data.

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